THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. An angle is 2 times its complement. The angle is: 1

(a) 70° **(b)** 55° **(c)** 65° **(d)** 60°

2. The smallest digit $*$ making $5*48$ divisible by 3 is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

3. The angles of a triangle are in the ratio $2:3:4$. The largest angle is: 1

(a) 100° **(b)** 80° **(c)** 70° **(d)** 90°

4. $1\frac{1}{2} \times 2\frac{1}{3}$ equals: 1

(a) $3\frac{1}{2}$ **(b)** $2\frac{1}{6}$ **(c)** $1\frac{3}{4}$ **(d)** 7

5. Half of the supplement of 40° is: 1

(a) 60° **(b)** 70° **(c)** 90° **(d)** 80°

6. The Collatz sequence from 8 (including 8 and 1) has how many terms? 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 3

7. A base angle of an isosceles triangle is 35°. The exterior angle at the apex is: 1

(a) 80° **(b)** 90° **(c)** 60° **(d)** 70°

8. $1\frac{1}{2} \div \frac{3}{4}$ equals: 1

(a) 2 **(b)** $1\frac{1}{8}$ **(c)** 4 **(d)** 1

9. Two interior angles on the same side of a transversal cutting parallel lines are in the ratio $1:2$. The smaller angle is: 1

(a) 80° **(b)** 50° **(c)** 70° **(d)** 60°

10. Applying Kaprekar's routine once to 4172 gives: 1

(a) 6175 **(b)** 6176 **(c)** 6173 **(d)** 6174

11. The exterior angle at B of $\triangle ABC$ is 120° and $\angle A = 45^\circ$. Then $\angle C$ is: 1

(a) 65° **(b)** 85° **(c)** 95° **(d)** 75°

12. $\frac{2}{3} \times \frac{3}{4} \div \frac{1}{2}$ equals: 1

(a) 1 **(b)** $\frac{1}{2}$ **(c)** $1\frac{1}{2}$ **(d)** 2

13. Four angles around a point are 75°, 105°, 90° and x°. Then $x =$ 1

(a) 80° **(b)** 110° **(c)** 90° **(d)** 100°

14. The number of 2-digit palindromes is: 1

(a) 11 **(b)** 10 **(c)** 8 **(d)** 9

15. Two sides of a triangle are 4 cm and 9 cm. The number of possible whole-number lengths of the third side is: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 9

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. An angle is 20° more than its complement. Find the angle. 2

17. The number $5 * 14$ is divisible by 9. Find the digit $*$ and write the number. 2

18. An isosceles triangle has perimeter 26 cm and one side 6 cm. Find its other sides (check which case is possible). 2

19. Simplify $\frac{2}{3} \times \frac{9}{10} \div \frac{3}{5}$. 2

20. Two angles forming a linear pair differ by 20°. Find both angles. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Two lines intersect and two adjacent angles are in the ratio $1 : 4$. Find all four angles. 3

22. Apply Kaprekar's routine to 7621 repeatedly until you reach 6174. 3

23. The angles of $\triangle ABC$ are in the ratio $2 : 3 : 4$. Find the angles and the exterior angle at the vertex of the largest angle. 3

24. A sheet is $1\frac{1}{2}$ m long and $2\frac{1}{4}$ m wide. Find its area and perimeter. 3

OR

An angle is 6° more than twice its complement. Find the angle and its complement. 3

25. The sum of a two-digit number and its reverse is 99, and its digits differ by 3. Find the number (tens digit larger). 3

OR

In $\triangle PQR$, $PQ = PR$, $\angle Q = (3x - 10)^\circ$ and $\angle R = (x + 6)^\circ$. Find x, all three angles and the exterior angle at P. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Two interior angles on the same side of a transversal cutting two parallel lines are in the ratio $2 : 3$. Find both angles and the angle forming a linear pair with the larger one. 5

OR

Write the Collatz sequence from 7. Find the largest number reached, the number of even terms and the number of odd terms. 5

27. The equal sides of an isosceles triangle are $(2x + 1)$ cm and $(3x - 4)$ cm and its base is $(x + 3)$ cm. Find x, the three sides and the perimeter. Check that the triangle exists. 5

OR

A wall of 240 m^2 is painted in three days. On day 1, $\frac{1}{4}$ of it is painted; on day 2, $\frac{2}{3}$ of the remainder; the rest on day 3. Find the area painted on each day. 5

END OF QUESTION PAPER **Total: 50 Marks**

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